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QUESTION 1

Write a program to take a list of 10 farmers, fruits they sell, their cities & the quantity of fruits they have as input. Save the data to notepad before & after sorting based on quantity in tabular manner. Also perform search operation based on farmers' name.

PROGRAM FOR THE QUESTION:

```
import java.io.*;
import java.util.*;

class Farmer {
   String name;
   String fruit;
   String city;
   int quantity;

   Farmer(String name, String fruit, String city, int quantity) {
      this.name = name;
      this.fruit = fruit;
      this.city = city;
      this.quantity = quantity;
   }
}
```

```
public class FarmerData {
   static void writeToFile(Farmer[] farmers, String filename) throws IOException {
      BufferedWriter bw = new BufferedWriter(new FileWriter(filename));
      String header = String.format("%-15s %-15s %-15s %-10s", "Name", "Fruit", "City", "Quantity");

 System.out.println("\n" + header);
bw.write(header);
      bw.newLine();
for (Farmer f : farmers) {
         String row = String.format("%-15s %-15s %-15s %-10d", f.name, f.fruit, f.city, f.quantity);
         System.out.println(row);
         bw.write(row);
         bw.newLine();
      }
      bw.close();
   }

   static void searchFarmer(Farmer[] farmers, String key) {
      boolean found = false; // Changed Boolean to boolean
      for (Farmer f : farmers) {
         if (f.name.equalsIgnoreCase(key)) { // Fixed 'id' to 'if'
            System.out.println("\nFarmer Found:");
            System.out.println("Name: " + f.name);
            System.out.println("Fruit: " + f.fruit);
            System.out.println("City: " + f.city);
            System.out.println("Quantity: " + f.quantity); // Fixed casing
            found = true;
```

```
            break; // Fixed quote to semicolon
        }
    }
    if (!found) {
        System.out.println("\nFarmer not Found.");
    }
  }
public static void main(String[] args) throws IOException {
    Scanner sc = new Scanner(System.in);
    Farmer[] farmers = new Farmer[10]; // Fixed 'Farmers' to 'Farmer'

    for (int i = 0; i < 10; i++) {
        System.out.println("\nEnter details for Farmer " + (i + 1));
        System.out.print("Name: ");
        String name = sc.nextLine();
        System.out.print("Fruit: ");
        String fruit = sc.nextLine();
        System.out.print("City: ");
        String city = sc.nextLine();
        System.out.print("Quantity: ");
        int quantity = sc.nextInt(); // Fixed data type
        sc.nextLine(); // Clear buffer

        farmers[i] = new Farmer(name, fruit, city, quantity);
    }
         System.out.println("Before Sort");
    writeToFile(farmers, "before_sort.txt"); // Fixed closing quote
```

```
    Arrays.sort(farmers, Comparator.comparingInt(f -> f.quantity));
        System.out.println("After Sort");
    writeToFile(farmers, "after_sort.txt");

    System.out.println("\nEnter farmer name to search:");
    String searchName = sc.nextLine();
    searchFarmer(farmers, searchName); // Fixed 'darmers' typo
    sc.close();
        }
}
```

OUTPUT:

```
c:\myJavaFile>javac FarmerData.java
C:\myJavaFile>java -cp . FarmerData

Enter details for Farmer 1
Name: F1
Fruit: APPLE
City: BSP
Quantity: 50

Enter details for Farmer 2
Name: F2
Fruit: BANANA
City: BSP
Quantity: 340
```

Enter details for Farmer 3

Name: F3

Fruit: ORANGE

City: RAIPUR

Quantity: 90

Enter details for Farmer 4

Name: F4

Fruit: LITCHI

City: BIHAR

Quantity: 300

Enter details for Farmer 5

Name: F5

Fruit: LITCHI

City: RAIGARH

Quantity: 200

Enter details for Farmer 6

Name: F6

Fruit: BANANA

City: EUROPE

Quantity: 140

Enter details for Farmer 7

Name: F7

Fruit: MANGO

City: BOTSWANA

Quantity: 250

Enter details for Farmer 8

Name: F8

Fruit: STRAWBERRY

City: FRANCE

Quantity: 450

Enter details for Farmer 9

Name: F9

Fruit: MELON

City: LADAKH

Quantity: 400

Enter details for Farmer 10

Name: F10

Fruit: MANGO

City: AFRICA

Quantity: 200

Name	Fruit	City	Quantity
F1	APPLE	BSP	50
F2	BANANA	BSP	340
F3	ORANGE	RAIPUR	90
F4	LITCHI	BIHAR	300
F5	LITCHI	RAIGARH	200

F6	BANANA	EUROPE	140
F7	MANGO	BOTSWANA	250
F8	STRAWBERRY	FRANCE	450
F9	MELON	LADAKH	400
F10	MANGO	AFRICA	200

Name	Fruit	City	Quantity
F1	APPLE	BSP	50
F3	ORANGE	RAIPUR	90
F6	BANANA	EUROPE	140
F5	LITCHI	RAIGARH	200
F10	MANGO	AFRICA	200
F7	MANGO	BOTSWANA	250
F4	LITCHI	BIHAR	300
F2	BANANA	BSP	340
F9	MELON	LADAKH	400
F8	STRAWBERRY	FRANCE	450

Enter farmer name to search:

F2

Farmer Found:

Name: F2

Fruit: BANANA

City: BSP

Quantity: 340

QUESTION 2

Write a program to take quantity & quality of fruits Mango, Apple & Papaya as input. Quality may be good, bad or average. If entered quantity of fruit is greater than 1000, store it as half. Perform search for a given fruit with given quality.

PROGRAM FOR THE CODE:

```
import java.util.*;


// Class to store fruit data
class Fruit {
   String name;
   String quality;
   int quantity;

   Fruit(String name, String quality, int quantity) {
      this.name = name;
      this.quality = quality;
      this.quantity = quantity;
   }
}

public class FruitData {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
```

```
// Fixed fruits
String[] fruitNames = {"Mango", "Apple", "Papaya"};
Fruit[] fruits = new Fruit[3];

// Input
for (int i = 0; i < 3; i++) {
    System.out.println("\nEnter details for " + fruitNames[i]);

    System.out.print("Quality (good/average/bad): ");
    String quality = sc.nextLine();

    System.out.print("Quantity: ");
    int quantity = sc.nextInt();
    sc.nextLine(); // clear buffer

    // If quantity > 1000, store half
    if (quantity > 1000) {
        quantity = quantity / 2;
    }

    fruits[i] = new Fruit(fruitNames[i], quality, quantity);
}

// Display stored data
System.out.println("\nStored Data:");
for (Fruit f : fruits) {
    System.out.println(f.name + " | " + f.quality + " | " + f.quantity);
```

```
}

// Search
System.out.println("\nEnter fruit name to search:");
String searchFruit = sc.nextLine();

System.out.println("Enter quality to search:");
String searchQuality = sc.nextLine();

boolean found = false;

for (Fruit f : fruits) {
    if (f.name.equalsIgnoreCase(searchFruit) &&
        f.quality.equalsIgnoreCase(searchQuality)) {

        System.out.println("\nFruit Found:");
        System.out.println("Name: " + f.name);
        System.out.println("Quality: " + f.quality);
        System.out.println("Quantity: " + f.quantity);

        found = true;
        break;
    }
}

if (!found) {
    System.out.println("\nFruit not found.");
```

```
        }

        sc.close();
    }
}
```

OUTPUT:

```
C:\myJavaFile>java -cp . FruitData

Enter details for Mango
Quality (good/average/bad): good
Quantity: 900

Enter details for Apple
Quality (good/average/bad): bad
Quantity: 300

Enter details for Papaya
Quality (good/average/bad): Average
Quantity: 1200

Stored Data:
Mango | good | 900
Apple | bad | 300
Papaya | Average | 600

Enter fruit name to search:
Apple
```

Enter quality to search:

bad

Fruit Found:

Name: Apple

Quality: bad

Quantity: 300

QUESTION 3

There are four orchards, namely A, B, C & D with fruits in them as Apple, Litchi, Mango, Apple respectively. The qualities of the fruits in the orchards are good, good, good & bad respectively.

The available orchards in a tree-view. Search for a given fruit in the available orchards after taking input from the user & print the tree.

PROGRAM FOR THE QUESTION:

```
import java.util.Scanner;

// Node class for tree structure
class OrchardNode {
    String orchardName;
    String fruit;
    String quality;
    OrchardNode left, right;

    OrchardNode(String orchardName, String fruit, String quality) {
        this.orchardName = orchardName;
        this.fruit = fruit;
        this.quality = quality;
        left = right = null;
    }
}

public class OrchardTree {
```

```
// Insert node level-wise manually for fixed 4 orchards
static OrchardNode createTree() {
    OrchardNode root = new OrchardNode("A", "Apple", "Good");
    root.left = new OrchardNode("B", "Litchi", "Good");
    root.right = new OrchardNode("C", "Mango", "Good");
    root.left.left = new OrchardNode("D", "Apple", "Bad");

    return root;
}

// Print tree in simple tree-view
static void printTree(OrchardNode node, int level) {
    if (node == null)
        return;

    for (int i = 0; i < level; i++)
        System.out.print("   ");

    System.out.println(node.orchardName + " -> " +
            node.fruit + " (" + node.quality + ")");

    printTree(node.left, level + 1);
    printTree(node.right, level + 1);
}

// Search fruit in tree
static void searchFruit(OrchardNode node, String key) {
    if (node == null)
        return;

    if (node.fruit.equalsIgnoreCase(key)) {
        System.out.println("Found in Orchard " +
                node.orchardName + " | Quality: " + node.quality);
    }

    searchFruit(node.left, key);
    searchFruit(node.right, key);
}
```

```
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        OrchardNode root = createTree();

        System.out.println("Available Orchards in Tree View:");
        printTree(root, 0);

        System.out.print("\nEnter fruit name to search: ");
        String fruitName = sc.nextLine();

        System.out.println("\nSearch Result:");
        searchFruit(root, fruitName);

        System.out.println("\nTree Structure:");
        printTree(root, 0);

        sc.close();
    }
}
```

OUTPUT:

```
c:\myJavaFile>javac OrchardTree.java

C:\myJavaFile>java -cp . OrchardTree

Available Orchards in Tree View:
A -> Apple (Good)
  B -> Litchi (Good)
    D -> Apple (Bad)
  C -> Mango (Good)

Enter fruit name to search: Apple

Search Result:
Found in Orchard A | Quality: Good
Found in Orchard D | Quality: Bad

Tree Structure:
A -> Apple (Good)
```

B -> Litchi (Good)
D -> Apple (Bad)
C -> Mango (Good)

QUESTION 4

Take a map provided as a graph & display all the subtrees of connected points.

PROGRAM FOR THE QUESTION:

```
import java.util.*;

// Graph using adjacency list
public class GraphSubtrees {

    static int vertices = 7; // Total points
    static ArrayList<Integer>[] graph = (ArrayList<Integer>[]) new ArrayList[vertices];
    static boolean[] visited = new boolean[vertices];

    // Add connection between two points
    static void addEdge(int u, int v) {
        graph[u].add(v);
        graph[v].add(u);
    }

    // DFS to print one connected subtree
    static void dfs(int node) {
        visited[node] = true;
        System.out.print(node + " ");
```

```
    for (int next : graph[node]) {
        if (!visited[next]) {
            dfs(next);
        }
    }
}

public static void main(String[] args) {

    // Initialize graph
    for (int i = 0; i < vertices; i++) {
        graph[i] = new ArrayList<>();
    }

    // Sample map connections
    addEdge(0, 1);
    addEdge(1, 2);

    addEdge(3, 4);

    addEdge(5, 6);

    System.out.println("All Subtrees of Connected Points:");

    // Find all connected components
    for (int i = 0; i < vertices; i++) {
        if (!visited[i]) {
```

```
            System.out.print("Subtree: ");
            dfs(i);
            System.out.println();
        }
    }
  }
}
```

OUTPUT:

```
C:\myJavaFile>javac GraphSubtrees.java
Note: GraphSubtrees.java uses unchecked or unsafe operations.
Note: Recompile with -Xlint:unchecked for details.

C:\myJavaFile>java -cp . GraphSubtrees
All Subtrees of Connected Points:
Subtree: 0 1 2
Subtree: 3 4
Subtree: 5 6
```

QUESTION 5

Enter entry point & goal point & produce the matching subtree of the graph.

PROGRAM FOR THE QUESTION:

```
import java.util.*;

// Find path (matching subtree) between entry point and goal point
public class GraphPath {

    static int vertices = 7;
    static ArrayList<Integer>[] graph = new ArrayList[vertices];
    static boolean[] visited = new boolean[vertices];
    static ArrayList<Integer> path = new ArrayList<>();
```

```
// Add edge between points
static void addEdge(int u, int v) {
    graph[u].add(v);
    graph[v].add(u);
}

// DFS to find path
static boolean findPath(int current, int goal) {
    visited[current] = true;
    path.add(current);

    if (current == goal)
        return true;

    for (int next : graph[current]) {
        if (!visited[next]) {
            if (findPath(next, goal))
                return true;
        }
    }

    // Backtrack if wrong path
    path.remove(path.size() - 1);
    return false;
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    // Initialize graph
    for (int i = 0; i < vertices; i++) {
        graph[i] = new ArrayList<>();
    }

    // Sample graph
    addEdge(0, 1);
    addEdge(1, 2);
    addEdge(1, 3);
    addEdge(3, 4);
```

```
        addEdge(4, 5);
        addEdge(5, 6);

        System.out.print("Enter Entry Point: ");
        int start = sc.nextInt();

        System.out.print("Enter Goal Point: ");
        int goal = sc.nextInt();

        if (findPath(start, goal)) {
            System.out.println("\nMatching Subtree / Path:");
            for (int node : path) {
                System.out.print(node + " ");
            }
        } else {
            System.out.println("\nNo Path Found.");
        }

        sc.close();
    }
}
```

OUTPUT:

```
C:\myJavFile>javac GraphPath.java
Note: GraphPath.java uses unchecked or unsafe operations.
Note: Recompile with -Xlint:unchecked for details.

C:\myJavaFile>java -cp . GraphPath
Enter Entry Point: 0
Enter Goal Point: 6

Matching Subtree / Path:
0    1 3 4 5 6
```

QUESTION 6

Give the weightage for each edge of nodes, and produce the shortest path from start to end node.

PROGRAM FOR THE QUESTION:

```
import java.util.*;
```

```
// Shortest path using Dijkstra Algorithm
public class ShortestPathGraph {

    static class Edge {
        int to, weight;

        Edge(int to, int weight) {
            this.to = to;
            this.weight = weight;
        }
    }

    static int vertices = 6;
    static ArrayList<Edge>[] graph = new ArrayList[vertices];

    // Add weighted edge
    static void addEdge(int u, int v, int w) {
        graph[u].add(new Edge(v, w));
        graph[v].add(new Edge(u, w));
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        for (int i = 0; i < vertices; i++) {
            graph[i] = new ArrayList<>();
        }

        // Sample weighted graph
        addEdge(0, 1, 4);
        addEdge(0, 2, 2);
        addEdge(1, 3, 5);
        addEdge(2, 1, 1);
        addEdge(2, 4, 8);
        addEdge(3, 5, 3);
        addEdge(4, 5, 2);

        System.out.print("Enter Start Node: ");
        int start = sc.nextInt();
```

```
System.out.print("Enter End Node: ");
int end = sc.nextInt();

int[] dist = new int[vertices];
int[] parent = new int[vertices];
boolean[] visited = new boolean[vertices];

Arrays.fill(dist, Integer.MAX_VALUE);
Arrays.fill(parent, -1);

dist[start] = 0;

// Dijkstra Algorithm
for (int i = 0; i < vertices - 1; i++) {
    int u = -1;
    int min = Integer.MAX_VALUE;

    for (int j = 0; j < vertices; j++) {
        if (!visited[j] && dist[j] < min) {
            min = dist[j];
            u = j;
        }
    }

    if (u == -1)
        break;

    visited[u] = true;

    for (Edge e : graph[u]) {
        if (!visited[e.to] && dist[u] + e.weight < dist[e.to]) {
            dist[e.to] = dist[u] + e.weight;
            parent[e.to] = u;
        }
    }
}

// Print shortest path
if (dist[end] == Integer.MAX_VALUE) {
```

```
            System.out.println("No Path Found.");
        } else {
            ArrayList<Integer> path = new ArrayList<>();
            for (int v = end; v != -1; v = parent[v]) {
                path.add(v);
            }

            Collections.reverse(path);

            System.out.println("\nShortest Path:");
            for (int node : path) {
                System.out.print(node + " ");
            }

            System.out.println("\nTotal Cost: " + dist[end]);
        }

        sc.close();
    }
}
```

OUTPUT:

```
c:\myJavaFile>javac ShortestPathGraph.java
C:\myJavaFile>java -cp . ShortestPathGraph
Enter Start Node: 0
Enter End Node: 5

Shortest Path:
0 2 1 3 5
Total Cost: 11
```

QUESTION 7

Write a program to demonstrate the use of all keywords made for exception handling.

PROGRAM FOR THE QUESTION:

```
import java.util.*;

public class ExceptionDemo {
```

```
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        try {
            System.out.print("Enter first number: ");
            int a = sc.nextInt();

            System.out.print("Enter second number: ");
            int b = sc.nextInt();

            int result = divide(a, b);   // may throw exception
            System.out.println("Result = " + result);
        }

        catch (ArithmeticException e) {
            System.out.println("Cannot divide by zero.");
        }

        catch (InputMismatchException e) {
            System.out.println("Invalid input. Enter integers only.");
        }

        catch (Exception e) {
            System.out.println("General Exception: " + e);
        }

        finally {
            System.out.println("Finally block executed.");
            sc.close();
        }
    }

    // Method using throws keyword
    static int divide(int x, int y) throws ArithmeticException {
        if (y == 0) {
            throw new ArithmeticException("Divide by zero");
        }
        return x / y;
    }
}
```

OUTPUT:

```
c:\myJavaFile>javac ExceptionDemo.java
c:\myJavaFile>java -cp . ExceptionDemo
Enter first number: 45
Enter second number: 63
Result = 0
Finally block executed.

c:\myJavaFile>java -cp . ExceptionDemo
Enter first number: 152
Enter second number: 0
Cannot divide by zero.
Finally block executed.

c:\myJavaFile>java -cp . ExceptionDemo
Enter first number: abc
Invalid input. Enter integers only.
Finally block executed.
```

QUESTION 8

Write a program to withraw money from a bank account, if the withdrawn money is greater than the balance, then it throws NoMoneyException.

PROGRAM FOR THE QUESTION:

```
import java.util.*;

// User-defined exception
class NoMoneyException extends Exception {
   NoMoneyException(String message) {
      super(message);
   }
}

public class BankWithdraw {

   static void withdraw(double balance, double amount) throws NoMoneyException {

      if (amount > balance) {
```

```
            throw new NoMoneyException("Insufficient Balance.");
        }

        balance = balance - amount;

        System.out.println("Withdrawal Successful.");
        System.out.println("Remaining Balance: " + balance);
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        try {
            System.out.print("Enter Account Balance: ");
            double balance = sc.nextDouble();

            System.out.print("Enter Withdrawal Amount: ");
            double amount = sc.nextDouble();

            withdraw(balance, amount);
        }

        catch (NoMoneyException e) {
            System.out.println("Exception: " + e.getMessage());
        }

        catch (Exception e) {
            System.out.println("Invalid Input.");
        }

        finally {
            sc.close();
        }
    }
}
```

OUTPUT:

```
C:\myJavaFile>javac BankWithdraw.java
```

```
C:\myJavaFile>java -cp . BankWithdraw
Enter Account Balance: 5000
Enter Withdrawal Amount: 1200
Withdrawal Successful.
Remaining Balance: 3800.0

C:\myJavaFile>java -cp . BankWithdraw
Enter Account Balance: 6000
Enter Withdrawal Amount: 12000
Exception: Insufficient Balance.

C:\myJavaFile>java -cp . BankWithdraw
Enter Account Balance: ab
Invalid Input.
```

QUESTION 9

Write a program which enables to use a platform by a train one at a time.

PROGRAM FOR THE QUESTION:

```
class Platform {

   // synchronized allows only one train at a time
   synchronized void usePlatform(String trainName) {

      System.out.println(trainName + " entered platform.");

      try {
         Thread.sleep(2000); // train stays for 2 seconds
      } catch (Exception e) {
      }

      System.out.println(trainName + " left platform.");
   }
}

// Train thread
class Train extends Thread {

   Platform p;
   String name;
```

```
    Train(Platform p, String name) {
        this.p = p;
        this.name = name;
    }

    public void run() {
        p.usePlatform(name);
    }
}

public class RailwayPlatform {

    public static void main(String[] args) {

        Platform p = new Platform();

        Train t1 = new Train(p, "Train A");
        Train t2 = new Train(p, "Train B");
        Train t3 = new Train(p, "Train C");

        t1.start();
        t2.start();
        t3.start();
    }
}
```

OUTPUT:

```
C:\myJavaFile>javac RailwayPlatform.java

C:\myJavaFile>java -cp . RailwayPlatform
Train B entered platform.
Train B left platform.
Train A entered platform.
Train A left platform.
Train C entered platform.
Train C left platform.

C:\myJavaFile>java -cp . RailwayPlatform
```

```
Train C entered platform.
Train C left platform.
Train B entered platform.
Train B left platform.
Train A entered platform.
Train A left platform.
```

QUESTION 10

Write a programme of Restaurant and Waiter, which accept order if kitchen if free and sheff deliver orders if he is free.

PROGRAM FOR THE QUESTION:

```
class Restaurant {

    boolean kitchenFree = true;
    boolean chefFree = true;

    // Waiter places order
    synchronized void placeOrder(String orderName) {

        if (kitchenFree) {
            kitchenFree = false;

            System.out.println("Waiter accepted order: " + orderName);
            System.out.println("Kitchen preparing " + orderName);

            try {
                Thread.sleep(2000);
            } catch (Exception e) {
            }

            kitchenFree = true;
        } else {
            System.out.println("Kitchen busy. Order rejected: " + orderName);
        }
    }

    // Chef delivers order
```

```
    synchronized void deliverOrder(String orderName) {

        if (chefFree) {
            chefFree = false;

            System.out.println("Chef delivered: " + orderName);

            try {
                Thread.sleep(1000);
            } catch (Exception e) {
            }

            chefFree = true;
        } else {
            System.out.println("Chef busy.");
        }
    }
}

// Waiter thread
class Waiter extends Thread {

    Restaurant r;
    String order;

    Waiter(Restaurant r, String order) {
        this.r = r;
        this.order = order;
    }

    public void run() {
        r.placeOrder(order);
    }
}

// Chef thread
class Chef extends Thread {

    Restaurant r;
    String order;
```

```
    Chef(Restaurant r, String order) {
        this.r = r;
        this.order = order;
    }

    public void run() {
        r.deliverOrder(order);
    }
}

public class RestaurantSystem {

    public static void main(String[] args) {

        Restaurant r = new Restaurant();

        Waiter w1 = new Waiter(r, "Pizza");
        Waiter w2 = new Waiter(r, "Burger");

        Chef c1 = new Chef(r, "Pizza");
        Chef c2 = new Chef(r, "Burger");

        w1.start();
        w2.start();

        c1.start();
        c2.start();
    }
}
```

OUTPUT:

```
C:\myJavaFile>javac RestaurantSystem.java
C:\myJavaFile>java -cp . RestaurantSystem
Waiter accepted order: Burger
Kitchen preparing Burger
Chef delivered: Burger
Chef delivered: Pizza
Waiter accepted order: Pizza
```

Kitchen preparing Pizza